

Exercises 2.5

In Exercises 1-6, solve the equation $Ax=b$ by using the LU factorization given for A. In Exercises 1 and 2, also solve $Ax=b$ by ordinary row reduction.

$$1. A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}, b = \begin{bmatrix} -7 \\ 5 \\ 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\boxed{\text{Solution:}} \quad L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

We need to solve the equations $Ly=b$ and $Ux=y$.
By Theorem 3 from Chapter 1, the matrix equation $Ly=b$ and $Ux=y$ has the same solution set as the system of equations corresponding to the augmented matrix $[L \ b]$ and $[U \ y]$.